

CHAMPS Contribution to the Third AIAA Ice Prediction Workshop

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Abstract

This paper presents the contribution of Polytechnique Montréal to the Third AIAA Ice Prediction Workshop (IPW3) using the in-house CHAMPS multiphysics software. The study provides a module-wise verification and validation of the numerical icing framework for the three-dimensional NACA0012 and ONERAM6 configurations, including aerodynamic flow, droplet impingement, thermodynamics, and ice accretion. Predictions are evaluated across the prescribed workshop grid hierarchies and droplet diameter classes droplet-size discretizations^{MA}. Results obtained Predictions^{MA} across the different modules of the icing framework remain consistent with grid and droplet-size discretization^{MA} refinements. Single-layer numerical^{MA} ice shapes leads to smooth ice contours with limited geometric complexity^{MA} within the experimental envelopes and exhibit limited sensitivity to the surface mesh resolutions spatial sensitivity^{MA}. Multilayer simulations produce more complex three-dimensional morphologies but do not necessarily improve agreement with experiment.

Introduction

Aircraft ice accretion is a multiphysics phenomenon involving aerodynamic flow, supercooled water droplet impingement, surface thermodynamics, and progressive evolution of the iced geometry^{MA}. Aircraft ice accretion^{MA} can substantially modify the effective shape of lifting surfaces and degrade aerodynamic performance [1, 2]. Reliable numerical icing tools therefore require accurate prediction not only of the final ice geometry but also of the individual physical components contributing to the accretion process like the computation of the aerodynamic flow, droplet trajectories and surface thermodynamic balance^{MA}.

Verification and validation are particularly important for numerical icing because numerical and modeling errors can propagate through successive components of the icing framework. Agreement in the final ice shape may result from compensating errors between these components. Evaluating aerodynamic, droplet-impingement, and thermodynamic quantities predictions^{MA} separately therefore helps identify the impact of each component on source of differences observed in^{MA} the final ice prediction.

The AIAA Ice Prediction Workshops provide configurations, computational grids, experimental measurements, and code-to-code comparisons to assess the state of the art in numerical icing simulation and identify differences among computational approaches [3, 4]. The Third

AIAA Ice Prediction Workshop (IPW3) extends this effort by emphasizing verification and validation through three-dimensional configurations with common grid hierarchies and experimental measurements enabling evaluation of spatial and droplet-size discretization effects^{MA}.

The present work summarizes the contribution of Polytechnique Montréal to IPW3 using the in-house CHAMPS multiphysics solver [5]. Two three-dimensional configurations are considered: the NACA0012 and the swept ONERAM6 wing. Two icing conditions, AE3932 and AE3933, are considered for the NACA0012, together with one icing condition for the ONERAM6. The corresponding icing conditions are summarized in Table 1.

Table 1: Icing conditions for the NACA 0012 and ONERA M6 test cases from the IPW3.

Condition	AE3932	AE3933	ONERA M6
M_∞	0.3151	0.3127	0.400
Re	4.4×10^6	4.4×10^6	4.2558×10^6
α (°)	4.0	4.0	2.0
T_∞ (K)	262.3	265.8	267.15
LWC (g/m ³)	0.55	0.55	0.51
MVD (μm)	20	20	20
Icing time (s)	420	420	1350

The contribution of this work is a component-wise verification and validation of the complete CHAMPS icing framework. Spatial refinement, droplet-size discretization using 1-, 3-, 7-, and 15-bin distributions^{MA}, Eulerian and Lagrangian impingement formulations, thermodynamics, and single- and multilayer ice shape solutions geometry evolution^{MA} are examined to identify the numerical and geometry modeling sensitivities influencing the final ice shape^{MA} prediction. Sensibility to the^{MA} spatial discretization is evaluated using a four-level grid hierarchy, ranging from coarse to fine hierarchies^{MA}. Levels are^{MA} prescribed for both configurations and are^{MA} summarized in Table 2.

Table 2: Grid hierarchy for the NACA0012 and ONERAM6 configurations, in millions of cells.

Grid level	NACA0012	ONERAM6
L4	5.86	0.135
L3	11.08	1.081
L2	26.99	8.651
L1	59.86	69.21

Methodology

The simulations are performed using the unstructured finite-volume CHAMPS multiphysics solver developed at Polytechnique Montréal [5]. The overall numerical icing sequence is illustrated in Fig. 1. The aerodynamic flow is first computed, followed by droplet trajectories impingement^{MA}, surface thermodynamics, and geometry evolution^{MA}. In a single-layer simulation, the complete exposure time is applied directly to a single accretion step. For multilayer simulations, the total exposure time is divided into N discrete^{MA} accretion steps that are performed sequentially. After each step, a new volume grid is generated^{MA} and the complete icing sequence is repeated for the updated geometry.

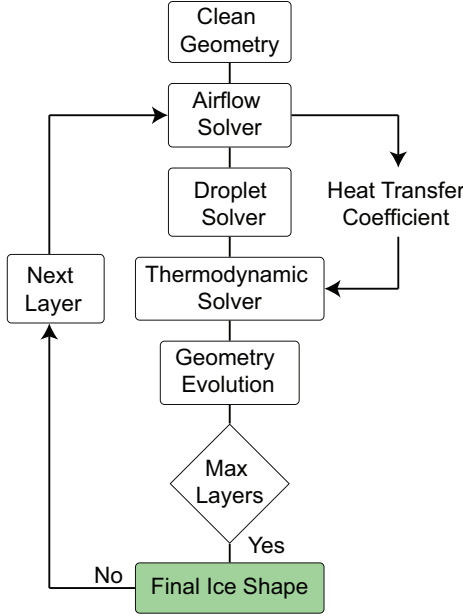


Figure 1: Multiphysics ice-accretion framework implemented in CHAMPS.

The aerodynamic calculations use the compressible Reynolds-averaged Navier-Stokes equations with the Spalart-Allmaras turbulence model [6] and the^{MA} Aupoix roughness correction [7]. A second-order Roe scheme [8] with the Venkatakrishnan limiter [9] is used for the convective flux discretization. Roe-convective fluxes [8] and a Venkatakrishnan limiter [9] are employed^{MA}.

Supercooled droplets are modeled using an Eulerian formulation based on the continuum representation of the dispersed phase and discretized using an upwind scheme [10, 11]. Complementary Lagrangian calculations are performed using successive refinement of the droplet-seeding cloud to ensure adequate convergence of the collected-water mass and independently verify the collection efficiency predicted by the Eulerian solver [12]. Droplet-size discretization is examined using 1-, 3-, 7-, and 15-bin representations of the experimental prescribed^{MA} droplet-size distribution. Each bin represents a discrete droplet diameter with a corresponding fraction of the total Liquid Water Content (LWC). allowing the influence of polydispersity on the impingement solution to be assessed.^{MA}

Surface thermodynamics are evaluated using an iterative formulation based on the classical Messinger surface energy balance [13]. Single-layer ice geometries are generated using an algebraic surface-displacement approach [11], while multilayer calculations use the continuous level-set methodology developed within CHAMPS. The level-set method provides an implicit representation of evolving interfaces through the zero contour of a scalar field [14, 15]. Following each accretion layer, the interface is extracted using MMG [16] and the aerodynamic domain is regenerated using automated POINTWISE procedures ref Pointwise?^{MA}.

Preliminary Results

NACA 0012

The AE3932 and AE3933 conditions differ primarily in freestream temperature, which has a negligible impact on the aerodynamic and droplet-impingement modules. These results are therefore discussed collectively as the NACA 0012 case, while the two conditions are distinguished when examining thermodynamics and ice accretion.

Using the grid hierarchy summarized in Table 2, the surface-pressure coefficient distributions remain consistent across grid levels, with differences primarily confined to the suction peak and stagnation region, as illustrated in Fig. 2.

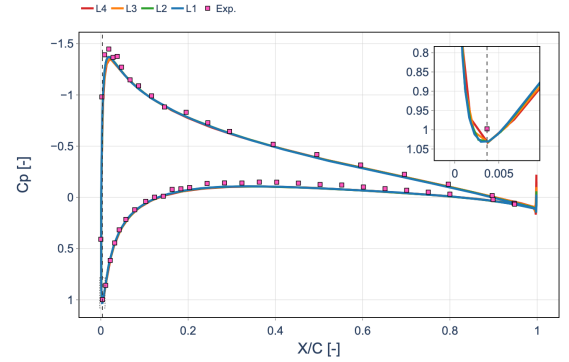


Figure 2: NACA0012 Level1 surface-pressure coefficient distribution.

Droplet impingement is examined through both spatial and droplet-size refinement. As shown in Fig. 3, increasing the number of droplet bins broadens the impingement region. The peak collection efficiency decreases slightly as the number of bins increases, while the largest differences occur in the impingement limits.

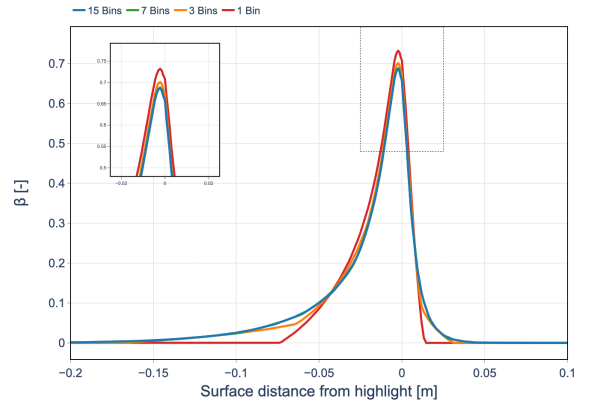


Figure 3: Effect of droplet-size discretization on the NACA 0012 collection-efficiency distribution.

Complementary Eulerian and Lagrangian calculations provide an independent verification of the impingement solution. On the Level1 grid, the difference in collected-water mass between the two formulations is less than 1% for both the single-bin and 7-bin distributions. The close agreement is maintained across the workshop grid hierarchy, as shown in Fig.4. Increasing the droplet representation from 1 to 7 bins increases the predicted collected-water mass by approximately 8%.

Experimentally, the ice mass decreases by approximately 15% between

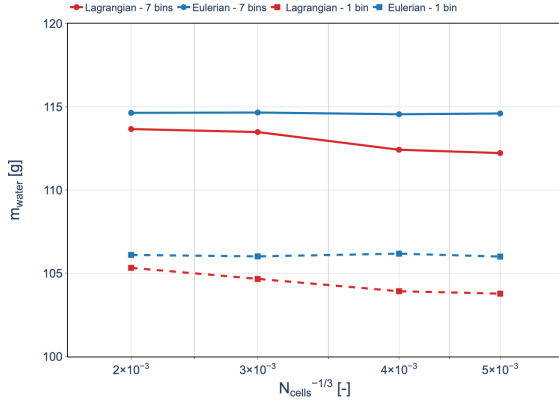


Figure 4: Eulerian and Lagrangian collected-water mass for the NACA 0012 using 1- and 7-bin droplet distributions.

case^{MA} AE3932 and case^{MA} AE3933. The numerical prediction exhibits a substantially smaller reduction of approximately 1.5%. This may be due to water shedding not being considered in the numerical model indicating insufficient thermodynamic sensitivity to the change in thermal condition^{MA}. Despite this discrepancy, the single-layer calculations reproduce the experimentally observed change in upper-horn orientation between the two conditions and exhibit limited sensitivity to the volume grid resolutionspatial sensitivity^{MA}, as shown in Fig. 5.

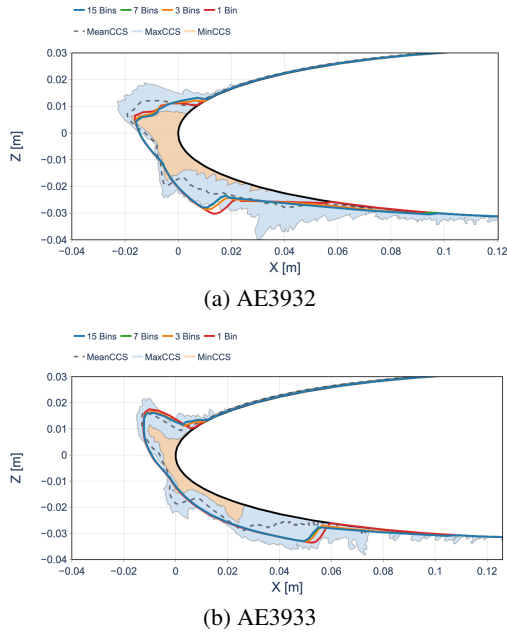


Figure 5: NACA 0012 Level 1 single-layer ice shapes showing the change in horn morphology between AE3932 and AE3933.

ONERA M6

The ONERAM6 configuration extends the verification and validation to a three-dimensional swept wing using the grid hierarchy summarized in Table 2. The aerodynamic predictions exhibit systematic grid spatial^{MA} convergence. As shown in Fig. 6, the surface-pressure distributions at the midspan stations^{MA}, $y = 0.75\text{m}$, show good consistency across the three finest grid levels, with the largest differences concentrated near the leading-edge suction peak. Level 4 exhibits the largest deviations, consistent with its comparatively coarse resolution of approximately 135,000 cells, compared with 5.86 million cells for the coarsest NACA0012 grid.^{MA}

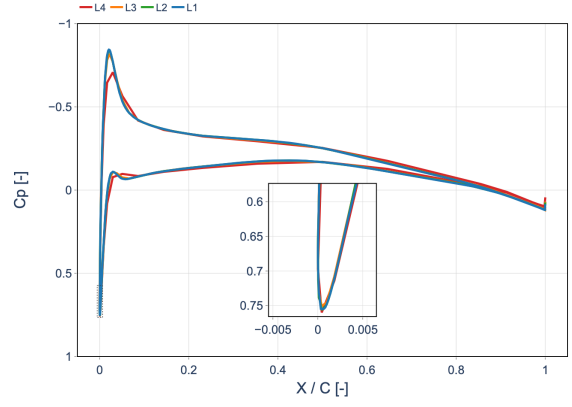


Figure 6: ONERA M6 surface-pressure coefficient distributions across the prescribed grid hierarchy at the midspan slice, $y = 0.75\text{m}$.

As withfor^{MA} the NACA0012 configuration, complementary Eulerian and Lagrangian droplet calculations produce closely matching collection-efficiency distributions for the ONERAM6. Fig. Figure^{MA} 7 shows the comparison at the $y = 0.75\text{m}$ intermediate^{MA} spanwise station on the Level 1 grid. Both formulations predict comparable peak collection efficiencies and impingement limits. The ONERAM6 droplet-size distribution is more closely centered around the Median Volumetric Diameter (MVD). This case has a narrower diameter range^{MA} from approximately 6 to $34\mu\text{m}$, whereas the NACA0012 diameter distribution extends to approximately $150\mu\text{m}$. Larger droplet diameters are more difficult to predict accurately and may require dedicated SLD models.^{MA}

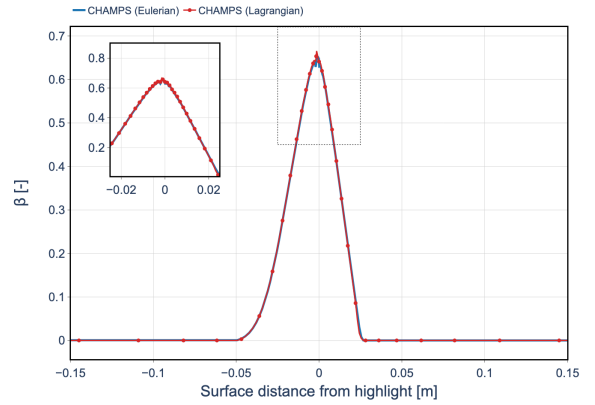


Figure 7: Eulerian and Lagrangian collection-efficiency distributions for the ONERA M6 on the Level 1 grid at $y = 0.75\text{m}$ using the single-bin droplet distribution.

Single-layer ice shapes exhibit relatively limited sensitivity to the volume mesh^{MA} over the finer workshop grids. Multilayer calculations using the continuous level-set framework produce pronounced three-dimensional horn structures and spanwise variations, as illustrated in Fig. 8.

Expected Results

The completed study will provide the full Polytechnique Montréal IPW3 verification and validation results for the NACA 0012 and ONERA M6 configurations. The final analysis will also investigate multilayer geometry evolution, including the influence of the choice of the direction of the surface displacementevolving surface normals^{MA}, the roughness model, and the ice-density model (an imposed value or a model?)modeling^{MA}. Comparisons with experimental measurements

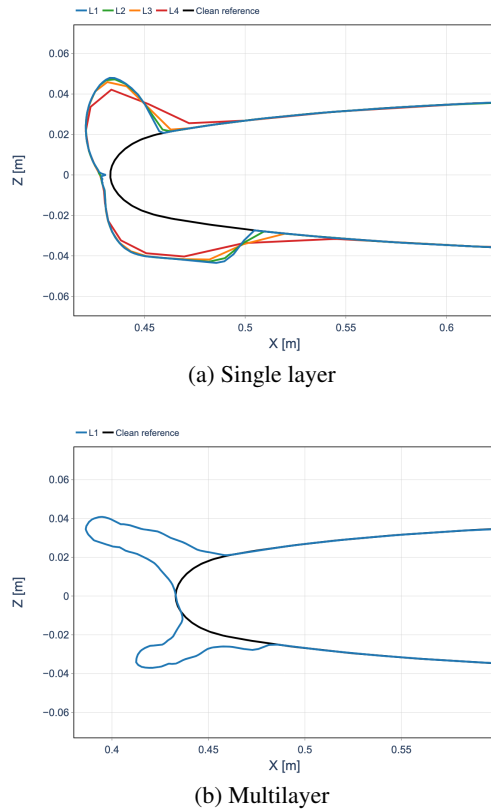


Figure 8: ONERA M6 Level 1 single-layer and multilayer ice-shape predictions at $y = 0.75$ m.

and IPW3 participant data will be used to determine how these modeling choices influence the integrated ice mass, horn orientation, and three-dimensional ice morphology.

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Definitions, Acronyms, Abbreviations

CFD	Computational Fluid Dynamics
IPW	AIAA Ice Prediction Workshop
LWC	Liquid Water Content
MVD	Median Volumetric Diameter
RANS	Reynolds-Averaged Navier–Stokes

General symbols

Greek symbols

C_p	Pressure coefficient	$[-]$
M_∞	Freestream Mach number	$[-]$
Re	Reynolds number	$[-]$
T_∞	Freestream temperature	$[K]$
y	Spanwise coordinate	$[m]$

α	Angle of attack	$[^\circ]$
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